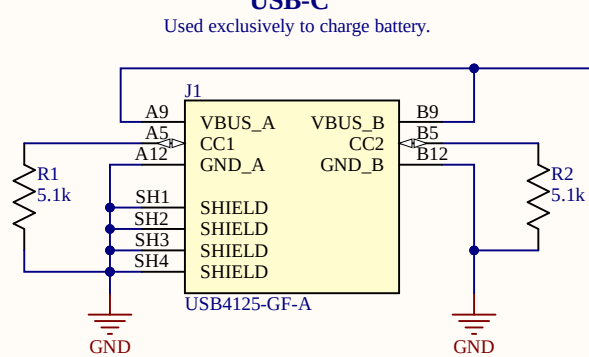


USB-C

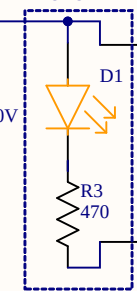
Used exclusively to charge battery.



USB-C plug will be placed on edge of the PCB for access through housing.

Power Charger

Charging Status

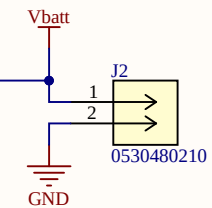


For best thermal performance, add vias from land area of EP to copper layer on opposite side of PCB

Power charger should be placed close to battery connector.

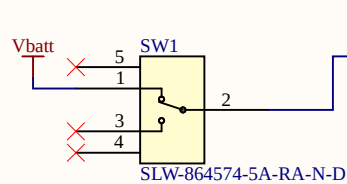
Li Ion Battery

LP384260JU+PCM+MOLEX 51021-0200 35MM



Connector will be placed on edge of the PCB to connect to battery located underneath.

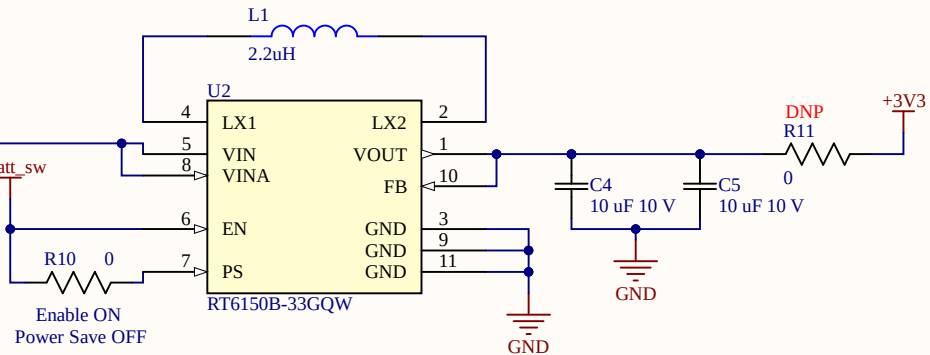
Power Switch



Switch in L position: ON
Switch in R position: OFF

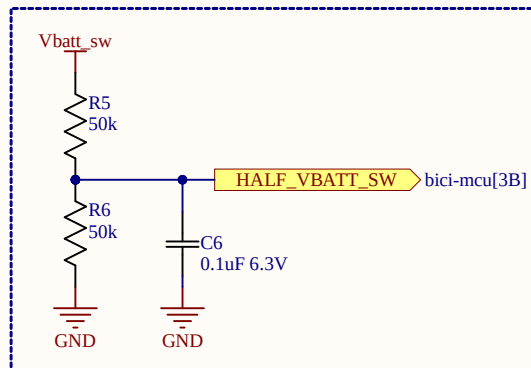
Switch will be placed on edge of the PCB for access through housing.

Buck-Boost

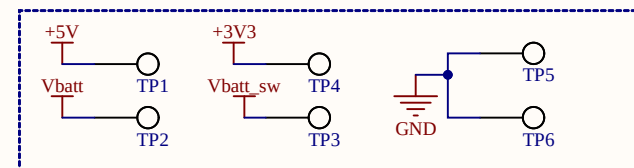


C3 should be placed as close as possible to Vin.
C4 and C5 should be placed as close as possible to Vout.
L1 should be connected to inductor by wide and short trace.

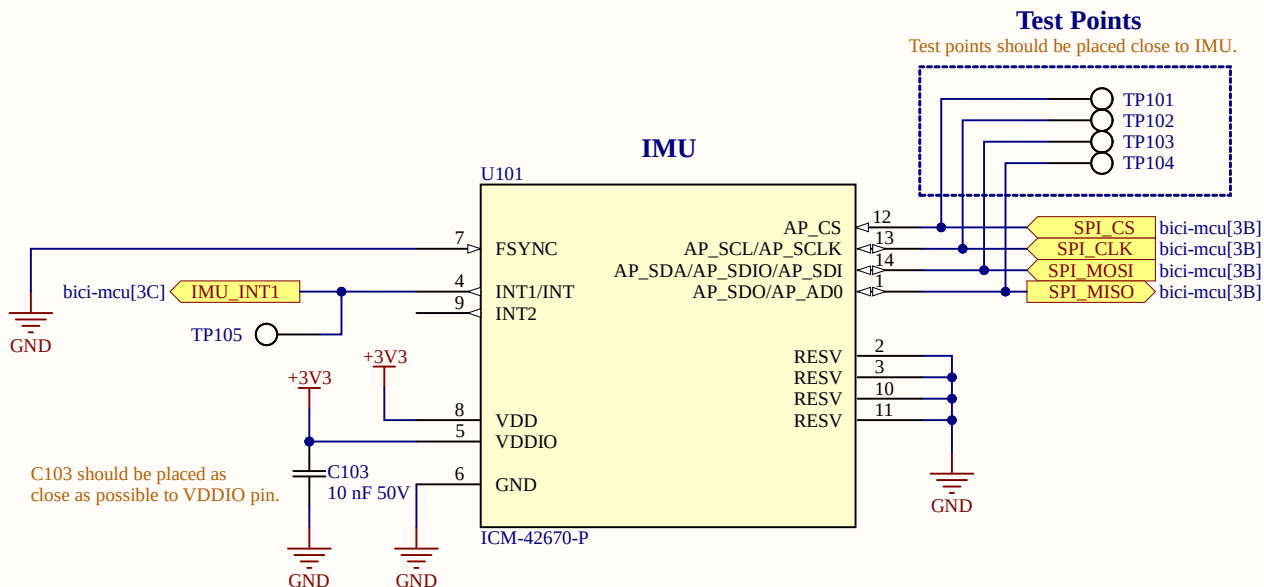
Low Power Detection



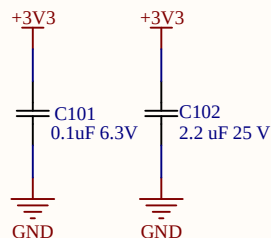
Test Points



Title		
Bici: Power Management		
Circuitry related to system power and charging.		
Size	Number	Revision
A		V5
Date:	2/24/2026	Sheet1 of 5
File:	C:\Users\...\bici-power.SchDoc	Drawn By: Team Bici



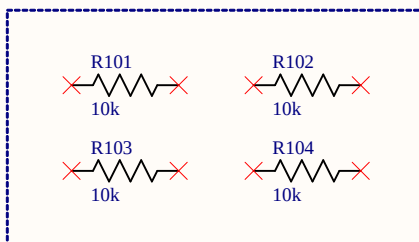
C103 should be placed as close as possible to VDDIO pin.



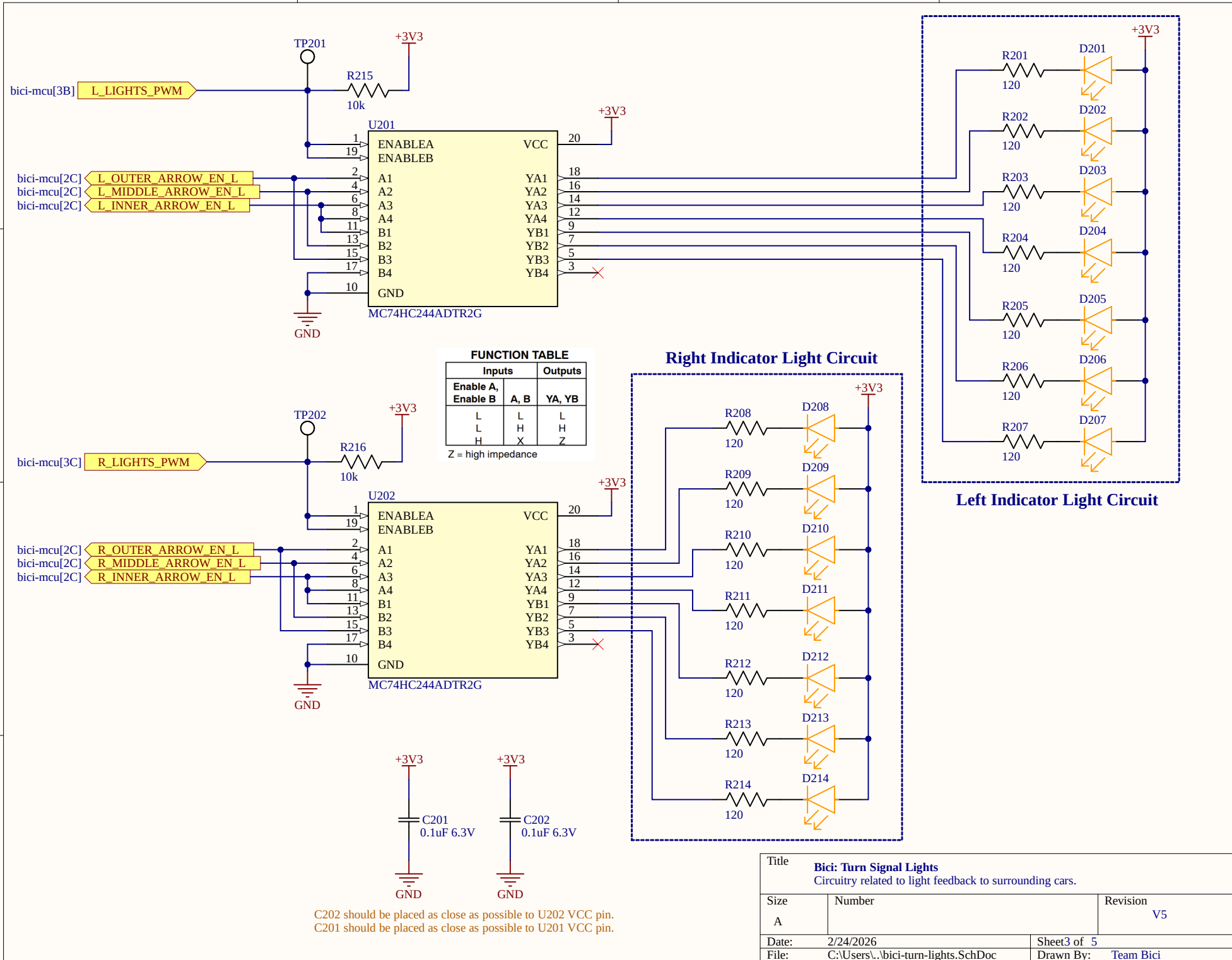
C101 and C102 should be placed as close as possible to VDD pin, with C101 closest to pin.

Spare 0603 Footprints

(Not on BOM) To make for easier soldering if we need to add a voltage divider (for eg) later.
(Per Mike Kofron's suggestion at Fall final presentation.)



Title Bici: IMU Circuitry related to IMU.		
Size A	Number	Revision V5
Date: 2/24/2026	Sheet2 of 5	
File: C:\Users\...\bici-imu.SchDoc	Drawn By: Team Bici	



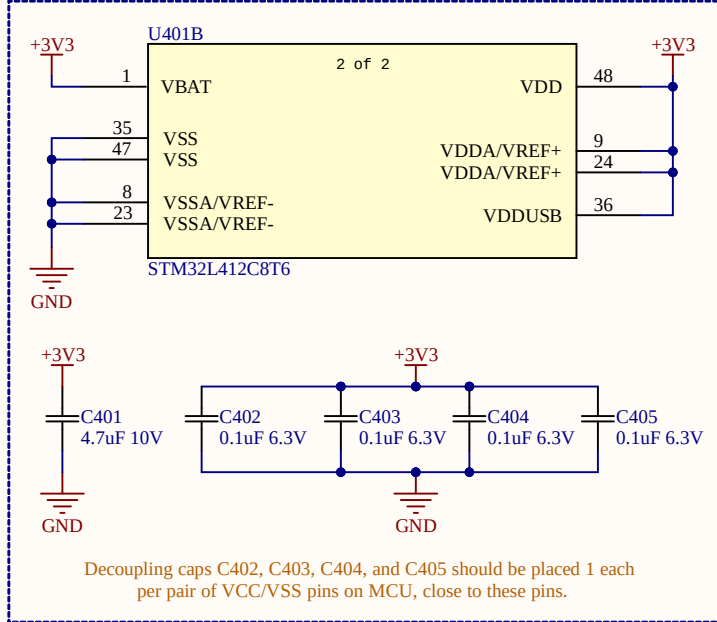
Title		
Bici: Turn Signal Lights		
Circuitry related to light feedback to surrounding cars.		
Size	Number	Revision
A		V5
Date:	2/24/2026	Sheet3 of 5
File:	C:\Users\...\bici-turn-lights.SchDoc	Drawn By: Team Bici

Right Indicator Audio Feedback Circuit

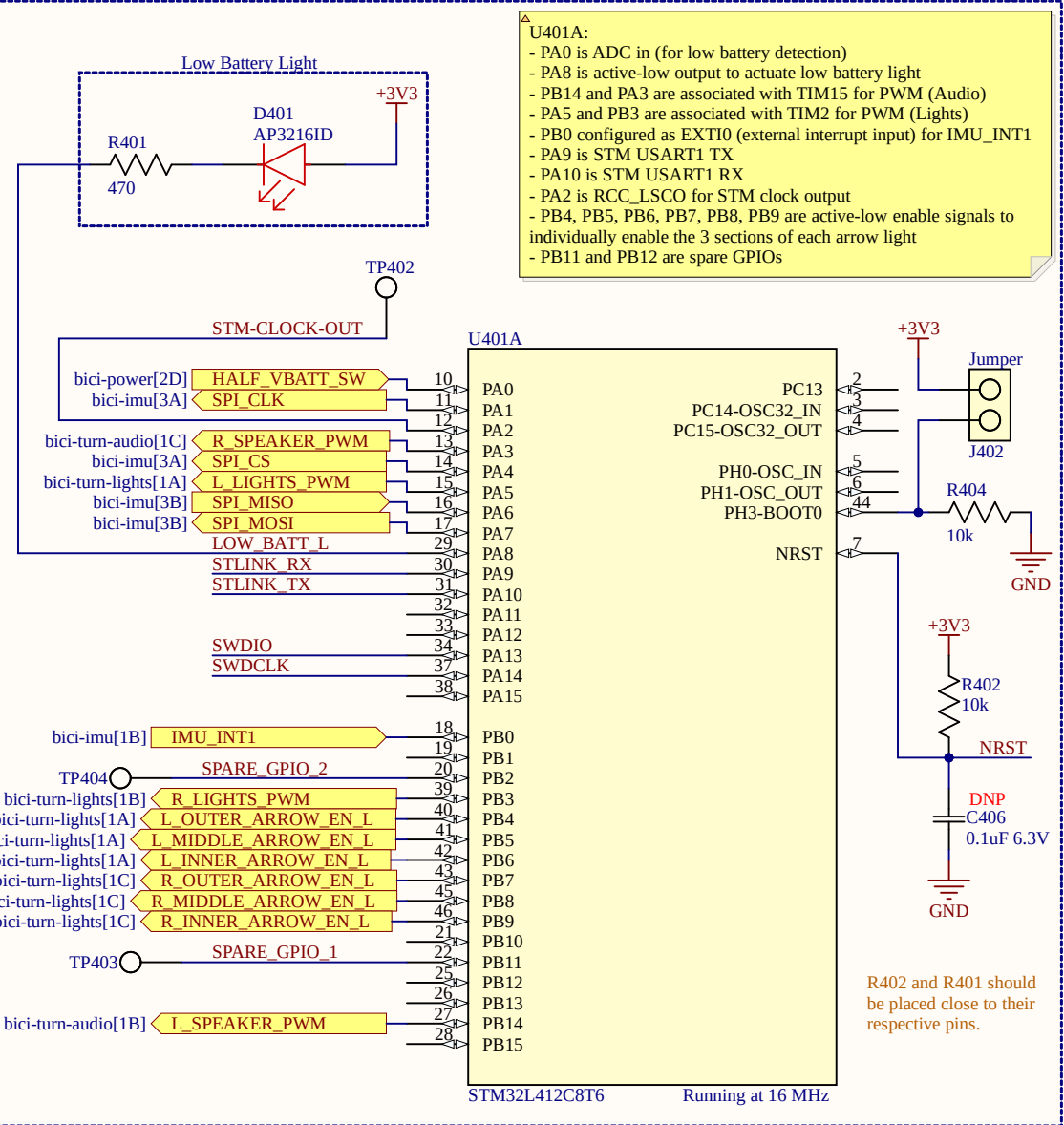
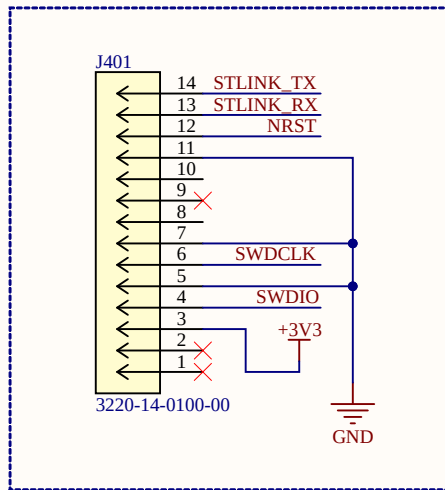
Title Bici: Turn Signal Audio Circuitry related to audio feedback to user.		
Size A	Number	Revision V5
Date: 2/24/2026	Sheet 4 of 5	
File: C:\Users\...bici-turn-audio.SchDoc	Drawn By: Team Bici	

STM Power

STM I/O



Programming/Debugging Connector



Title			Bici: MCU	
			Circuitry related to embedded MCU.	
Size	Number		Revision	
A			V5	
Date:	2/24/2026		Sheet5 of 5	
File:	C:\Users\...\bici-mcu.SchDoc		Drawn By: Team Bici	

